

Vihanga Rathnayake

INFORMATION SYSTEMS UNDERGRADUATE | BUSINESS ANALYSIS | ERP & DIGITAL TRANSFORMATION

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Summary

Final-year Information Systems undergraduate with expertise in enterprise systems, software development, business analysis, and digital transformation. Experienced in developing scalable applications, modeling business processes, and translating organizational requirements into effective technology solutions. Skilled in cloud technologies, requirements engineering, BPMN, UML, and ERP concepts, with a passion for driving process optimization and technology-enabled organizational growth.

Education

Higher National Diploma in Software Engineering	<i>October 2024 – Present</i>
Faculty of Engineering Technology, Open University of Sri Lanka Second Year	CGPA: 3.54
Bachelor of Computing (Hons.) in Information Systems	<i>June 2023 – Present</i>
Faculty of Computing, University of Sri Jayewardenepura Final Year	CGPA: 3.18
Assured Diploma in Information Technology	<i>Completed 2022</i>
Esoft Metro College, Monaragala	Results: Distinction
G.C.E. Advanced Level Examination (Physical Science Stream)	<i>Completed 2021</i>
Dutugemunu College, Buttala	Results: BCC
Vadhya Visharad (Tabla)	<i>Completed 2017</i>
Bhatkhande Sangeet Vidyapith, Lucknow	Second Upper Class

Project Experience

ML-Driven Geotechnical Report Generation Platform – Final Year Research Project *2026 (Ongoing)*

- Developing an AI-powered platform to automate Low Strain Pile Integrity Test (PIT) analysis and report generation, reducing manual effort, and improving consistency in geotechnical reporting.
- Designed a microservice-based system integrating data parsing, signal feature extraction, and machine learning models trained on 22,000+ Sri Lankan bored pile records to classify pile integrity and generate ASTM D5882-compliant engineering reports through a React-based application.

Technologies: Python, Fast API, Scikit-learn (Random Forest, SVM), React.js, PostgreSQL, REST

PrimeHeal – Smart Healthcare Support System - Group Project *2026 (Ongoing)*

- Serving as the Database Administrator, responsible for designing and maintaining the database infrastructure supporting patient records, appointments, and user operations.
- Designed and optimized relational database schemas, queries, indexing strategies, and normalization structures to ensure data integrity, security, scalability, and high-performance access to healthcare information.

Technologies: MySQL, React.js, Node.js, Express.js, JWT Authentication, Tailwind CSS

ERP Implementation Business Case for PiLETesT Consultants (Pvt) Ltd *2026 (Ongoing)*

- Conducted an enterprise systems assessment to identify operational inefficiencies and digital transformation opportunities across project management, procurement, inventory, equipment management, and financial operations.
- Analyzed business requirements and developed an Odoo 18 ERP implementation roadmap, including BPMN process models and functional specifications to support workflow integration, process automation, ISO 9001 compliance, and data-driven decision-making.

Technologies: Odoo 18 ERP, BPMN, Business Process Analysis, Requirements Engineering, Enterprise Systems

Personal Portfolio Website *2026*

- Designed and deployed a full-stack personal portfolio using React.js, TypeScript, Vite, and Tailwind CSS featuring a skills radar chart (Recharts), particle hero section, cinematic project gallery, digital business card, and an AI-powered chatbot integrated via the Claude API.
- Deployed via GitHub Pages with automated gh-pages CI/CD pipeline.

Technologies: React.js, TypeScript, Vite, Tailwind CSS, Recharts, Claude API, GitHub Pages, CI/CD

[Visit Projects](#)

Certifications

Networking Basics
Data Science
Google Analytics

Cisco
PyData
Google

Publications

"Digital Transformation in Geotechnical Engineering: A Systematic Review of Mobile Systems and AI-Driven Report Generation," *1st Jayewardenepura Computing Symposium*, University of Sri Jayewardenepura, 2026.

Skills & Interests

Business Analysis: Requirement Engineering, BPMN, Process Mapping, ERP
Programing: Python, Java, C#, C++
Databases: SQL & NoSQL DBs (MySQL, MongoDB)
DevOps/Cloud: GitHub Actions, AWS, CI/CD
Tools: Git, GitHub, Figma, VS Code, IntelliJ IDEA
Soft Skills: Teamwork, Problem Solving, Fast Learner, Ability to adapt quickly, Self-motivated.
Interests: Cloud, AI tools, Automation, Digital Transformation

Volunteering

Flair club, University of Sri Jayewardenepura

Co-Secretary 2025-2026
Oversaw and managed all club documentation and reporting processes. Supervised club's projects.

Faculty Coordinator, Faculty of Computing 2023-2024
Facilitated communication between faculty students and club leadership.

Literature Sub Committee of Students Union, University of Sri Jayewardenepura

President 2025-2026
Led the planning and coordination of literary and cultural initiatives to promote student engagement and creative expression within the university community.

Association of Computing Students, University of Sri Jayewardenepura

Co-Founder & Secretary 2024 – 2025
Co-founded the faculty's official subject association and coordinated executive operations, stakeholder engagement, and academic development initiatives.

Founder & Chair of project EduACS 2024 – 2025
Founded and led an educational outreach initiative, coordinating GCE(A/L) IT and Math seminars, stakeholder partnerships, and project teams to support secondary school students.

Sasnaka Sansada Foundation

Chair of Project Poyata Padamak 2022-2023
Led a recurring GCE (O/L) online mathematics seminar initiative conducted on every Poya day, coordinating speakers, volunteers, and program delivery for student learning and engagement.

Representative & Ambassador Roles

University Ambassador, G17 Global 2024 -2025
University Ambassador, INSL 2025-2026
University Student Ambassador, National Child Protection Authority 2024 –2025
Liaison Officer, International Solar Alliance 2024-2025

Honors & Awards

Logistics & Facilitation Excellence Award, Flair Club of University of Sri Jayewardenepura June 2025

Ambassador of the Month, G17 Global July 2024

Presidential Award, Central Environmental Authority November 2017

References

Dr. Nuwan Jayarathne

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